Ex. No: 6

Date: 10/9/26

Error correction at Data Link layer using Hamming code.

AIM:

Write a program to implement error detection and correction using Hamming code concept. make a test run to input data stream and verify error correction feature.

Create sender program with below features.

1. Input to sender file should be a text of any length. program should convert the text to binary.

2. Apply hamming code concept on the binary data and add redudent bits to it.

3. Save this output in a file called channel.

Create a receives program with below features.

1. receiver program should read the input from channel file.

2. Apply hamming code on the binary data to check for errors.

3. if there is an error, display the position of error

4. Else remove the redundant bits and convert the binary data to Ascii and display the output

Code:

```
def hamming_encryption(data):
    m = len(data)
    r = 0
    while (2**r) <= (m+r+1):
        r += 1
    r -= 1
    n = m+r+1
    code = [0] * n
    j = 0
    for i in range(1, n):
        if (i & (i-1)) != 0:
            code[i] = data[j]
            j += 1
    for i in range(r):
        pos = 2**i
        pair = 0
        for j in range(1, n):
            if j & pos:
                pair ^= int(code[j])
        code[pos] = str(pair)
    return (code[1])

def hammingcode_deseryption(data, r, m):
    for i in range(r):
        pos = 2**i
        pair = 0
```

```
for j in range (1, r+m+1):
    if j & pos:
        pair ^= int (code[j])
if (pair != 0):
    if (pos == '0'):
        data [pos] = '1'
    else
        data [pos] = '0'
return join (data)
data = input ("enter binary data")
print ("Data after Encoding from sender side",
    hamming code - encryption)
rdata = input ("Enter data received by the reciever")
print ( hammig code - decryption (rdata, r, rdata length))
```

frame / received

Circular Queue between senders & receiver

output:

Enter the data : 1101

No. of recurrent bit is : 3

Data transimitted from sender : 1010101

Index you want to chage : 6

Data received from received : 1010111

After performing error detection & correction 1010101

Result:

thus the hamming code was implemented sucessfully for error detection and correction.